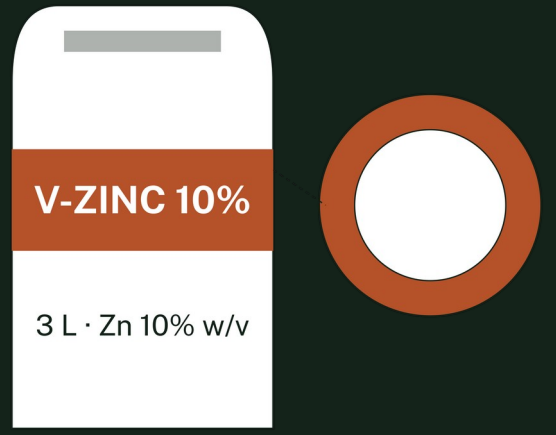
DEALER TRAINING · NOVEMBER 2026

V-Zinc 10%



Liquid zinc sulfate, 100 g of zinc in a litre, applied in the irrigation water or sprayed on the leaf at early growth.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

V-Zinc 10% is a liquid zinc sulfate with 10% zinc, weight in volume: 100 g of zinc in every litre. It dissolves fully and leaves no residue in the tank or the emitter. The plant needs zinc to make the growth hormone that lengthens the stem and the enzymes that build protein, so zinc controls plant height, leaf size and, in rice and wheat, grain set. On calcareous soil above pH 8, zinc is held as zinc phosphate or on the lime, in forms the root cannot take up. In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), 59.0% of samples tested below 1.0 ppm zinc. V-Zinc is therefore applied as a solution, on the leaf or in the irrigation water, at early growth, at the rate in the crop plan. It goes on early, while the plant's size is being set. V-Zinc is VAN's liquid zinc; VAN's granular zinc for the soil at sowing is V-Transform.

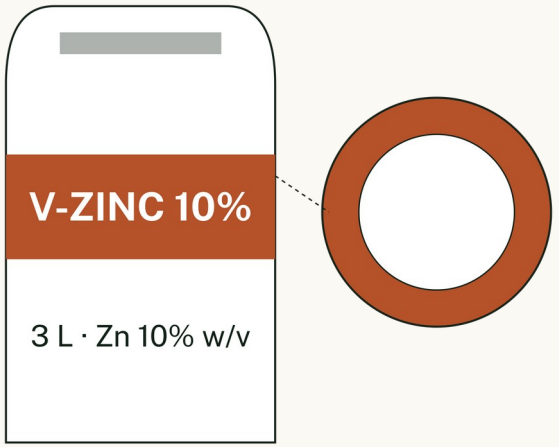
The 4 points it must cover

1. The pack: 3 L, Zn 10% w/v as zinc sulfate, PS 5469-2020. 300 g of zinc in the pack, 100 g in every litre. 20 L and 200 L packs are on the price sheet.
2. The nutrient: the growth hormone, leaf size, protein and grain set. 59.0% of Punjab soil samples (about 2016 to 2018) tested below 1.0 ppm zinc.
3. The stage: early growth, on the leaf or in the irrigation water, at the rate in the crop plan.
4. The rule: never in the same tank as a soluble phosphate. Zinc phosphate does not dissolve.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the pack, in the published crop plan or in the named evidence on the slides that follow.

THE PACK

What the pack says, and what it is registered as.



3 L

the pack every crop plan row is written in. 300 g of zinc in it, from the declared analysis. 20 L and 200 L packs are on the price sheet.

Zn 10%

zinc, w/v: 100 g in every litre, as zinc sulfate. The crop plan rows give it as "Zinc 10% w/v".

Sulfate

The zinc is in the form of zinc sulfate, as registered. It dissolves at once in water and is taken up through the leaf when sprayed.

2 routes

spray on 3 rows and fertigation on 1, at early growth, as the crop plans give them.

59.0%

of 770,160 Punjab soil samples (about 2016 to 2018) tested below 1.0 ppm zinc. The mean was 1.25 ppm; samples with a higher pH had less zinc.

4

published crop plans name V-Zinc, in 4 rows, all written in the 3 L pack.

Registered as	Pakistan Standard	Manufacturing licence
V-Zinc 10%	PS 5469-2020	CM/L-3130/2024

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“The rice nursery has brown spots and the plants stay short. In the maize a white band runs down the young leaf. I gave both their urea. What is it?”

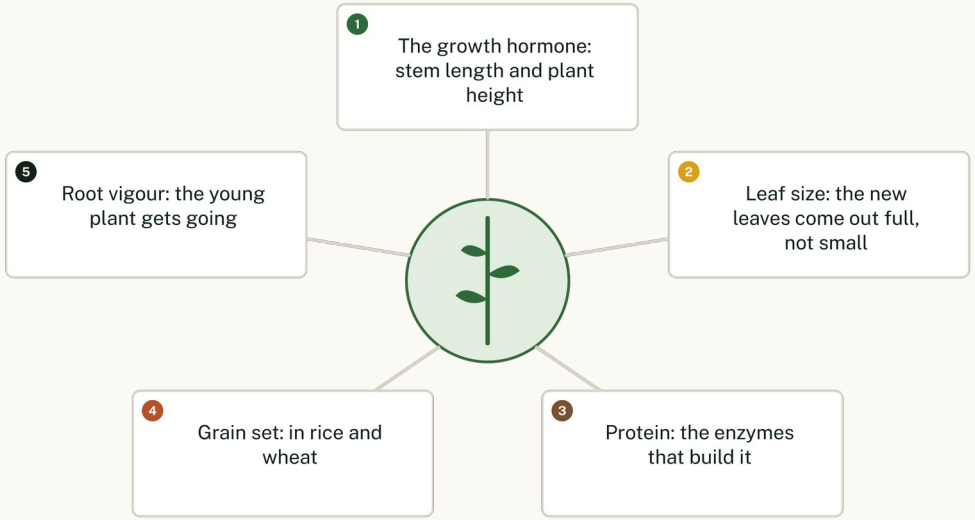
The grower is describing zinc shortage. Rice and maize show it more clearly than most crops, especially on calcareous soil. He applied urea, but the crop cannot use that nitrogen well without enough zinc, the micronutrient that controls growth. The answer has 3 parts: what zinc does, why calcareous soil leaves the crop short of it, and why a solution at early growth reaches the crop.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

Zinc controls growth, and a zinc-short crop stays short, with small, narrow young leaves.

The plant needs zinc to make the growth hormone that lengthens the stem and the enzymes that build protein. Zinc therefore controls plant height, leaf size and, in rice and wheat, grain set. A crop short of zinc has small, narrow young leaves that are yellow between the veins, and short joints that make the plant look bunched. Young maize leaves show a white band; older rice leaves in the nursery show brown spots. A crop can stop growing for lack of zinc even in a field with plenty of nitrogen.



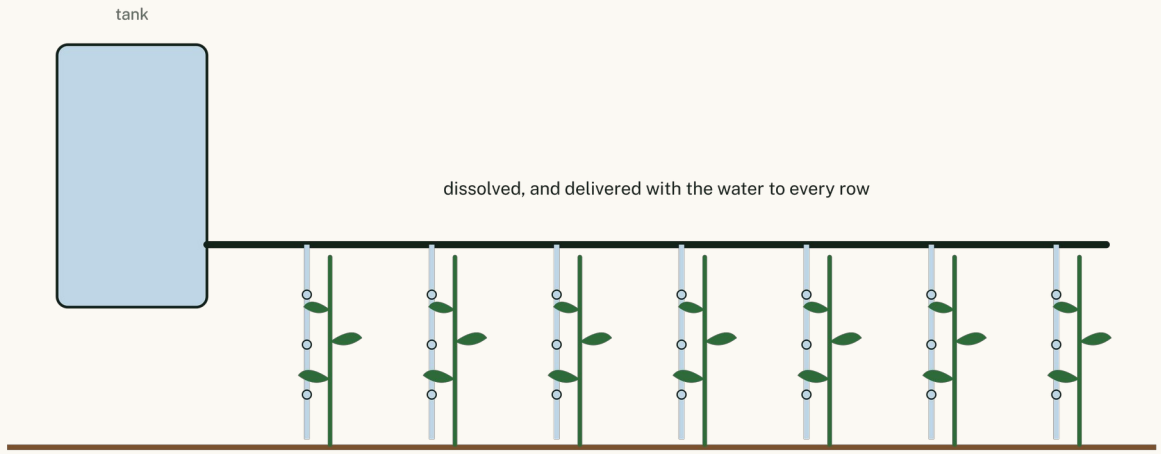
The jobs of zinc, from VAN's 17-nutrients page. The drawing shows the idea; it is not a measurement.

Say it like this: "Zinc helps the plant grow tall and its leaves grow to full size. Without enough zinc the crop stays small, however much urea you give it."

ANSWER 2 OF 3

The soil holds zinc but locks it, most of all above pH 8.

On calcareous soil above pH 8, zinc is fixed as zinc phosphate or held on the calcium carbonate, so the root cannot take it up even when the soil holds enough in total. In the Punjab soil survey (about 2016 to 2018), 69.1% of samples had a pH above 8 and 59.0% tested below 1.0 ppm zinc. Samples with a higher pH had less zinc, and samples with more than 3% lime had much less. Decades of phosphate fertiliser on these soils make zinc shortage worse, because zinc and phosphorus interfere with each other. A solution put in with the irrigation water reaches the wet root zone as zinc the root can take up, and a spray reaches the leaf directly, without passing through the soil.



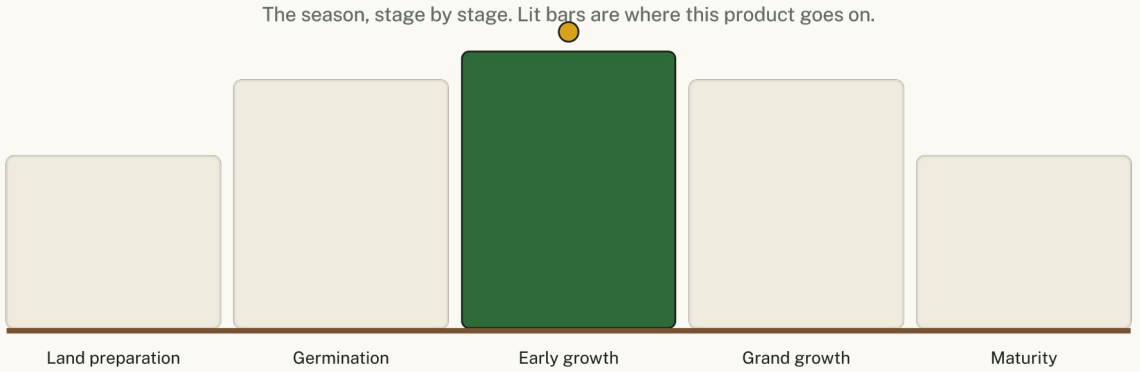
The fertigation line shows a solution in the wet root zone. The slide uses general soil chemistry and the Punjab soil survey, not a product measurement.

Say it like this: “Your soil has zinc, but the soil holds it in a form the root cannot take up. So we give zinc as a liquid, in the water or on the leaf.”

ANSWER 3 OF 3

Apply V-Zinc early, while the plant’s final size is still being set.

The enzymes of early growth, and the hormone that sets how tall and wide the plant grows, both need zinc. The crop plans put V-Zinc on early in the season: ⅕ pack by spray at early growth on chickpea, lentil and mungbean and mash; 2 packs by fertigation at early growth on mango. Zinc applied at flowering comes too late to change the plant’s size, because the shortage has already limited it. Apply it early, at the stage the crop plan gives.



The 5 stage names on the drawing are the ones VAN’s crop plans use. The crop plan slides show where V-Zinc goes for each crop.

Say it like this: “Give the zinc early, while the plant’s size is being set. Zinc given late reaches a plant whose size is already set.”

THE EVIDENCE

What is verified for V-Zinc 10%, and what is not.

No trial of V-Zinc is on VAN's record. A dealer may quote the analysis on the pack, the licence, the crop plan rows and the Punjab soil survey, each only as the right-hand column allows.

Item	What it says	How a dealer may use it
The pack	V-Zinc 10%, zinc sulfate, Zn 10% w/v, 100 g per litre, liquid, fully soluble. 3 L; 20 L and 200 L on the price sheet.	As fact. Grams per litre from the declared analysis.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-3130/2024, as the master record holds it.	As fact: quote the number.
The crop plan rows	4 rows in 4 crop plans, all in the 3 L pack: ⅕ pack by spray at early growth on chickpea, lentil and mungbean and mash; 2 packs by fertigation at early growth on mango.	As the published rate, word for word.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	59.0% of samples below 1.0 ppm zinc; mean 1.25 ppm; less zinc at higher pH (1.45 to 1.20 ppm) and above 3% calcium carbonate. 69.1% above pH 8.	As fact about the soil. Never as product performance.
The chemistry (textbook)	Above pH 8 zinc is fixed as zinc phosphate or on lime; zinc and a soluble phosphate in 1 tank make zinc phosphate, which does not dissolve. Zinc and phosphorus interfere with each other in the soil.	As general soil chemistry, not a VAN measurement.
Not on file	A V-Zinc trial; a yield or grain-zinc figure; the 64% and 57% soil figures and the zinc findings of the National Nutrition Survey printed in earlier VAN material; a 1 L foliar pack with iron, amino acids and seaweed; a drone rate; a chelate.	Not said until VAN names a source.

Sources: the pack; VAN's published crop plans on van.com.pk; the farmer price sheet, Aug 2026; the PSQCA master record; the Punjab soil survey, 770,160 samples tested in about 2016 to 2018; VAN's 17-nutrients page; VAN's own product data.

The 3 packs, in grams of zinc and in rupees.

Prices are the delivered farmer prices from the VAN farmer price sheet, Aug 2026. Zinc is from the declared analysis, weight in volume: 10% is 100 g in a litre.

V-Zinc 10%, Zn 10% w/v	Volume	Zinc	Rupees	Rupees per litre
1 pack	3 L	300 g	1,375	458
1 pack	20 L	2.0 kg	6,450	323
1 drum	200 L	20 kg	55,000	275

A 3 L pack is 300 g of zinc for Rs 1,375. In the 20 L pack a litre costs Rs 323, against Rs 458 in the 3 L pack. The largest crop plan row is mango: 2 packs of 3 L, 6 litres an acre, for Rs 2,750. The 20 L pack costs less per litre and suits a grower dosing many acres.

V-Zinc 3 L Rs 1,375, 20 L Rs 6,450, 200 L Rs 55,000: the farmer price sheet, Aug 2026. 300 g = 3 × 100; 2.0 kg = 20 × 0.1. Rupees per litre = price ÷ litres. 2 × 3 L = 6 L; 2 × 1,375 = 2,750.

What the crop plans apply per acre, in grams of zinc.

Packs per acre are from the published crop plans, in the 3 L pack; zinc is from the declared analysis and rupees from the VAN farmer price sheet, Aug 2026.

Crop	Stage and method, as printed	3 L packs	Litres	Zinc	Rupees per acre
Chickpea	Early Growth, spray	⅕	0.6	60 g	275
Lentil	Early Growth, spray	⅕	0.6	60 g	275
Mungbean & Mash	Early Growth, spray	⅕	0.6	60 g	275
Mango	Early Growth, fertigation	2	6	600 g	2,750

The spray rows apply zinc by the gram: an acre of chickpea receives 60 g of zinc as a leaf spray, costing Rs 275. Mango takes 600 g in the irrigation water. The table shows the same pack used 2 ways: a dose in the irrigation water to the root zone, and a small dose sprayed on the leaf.

VAN's published crop plans on van.com.pk, rows as printed in the 3 L pack. Rupees = packs × 1,375, farmer price sheet, Aug 2026.

The liquid and the granule do 2 different jobs.

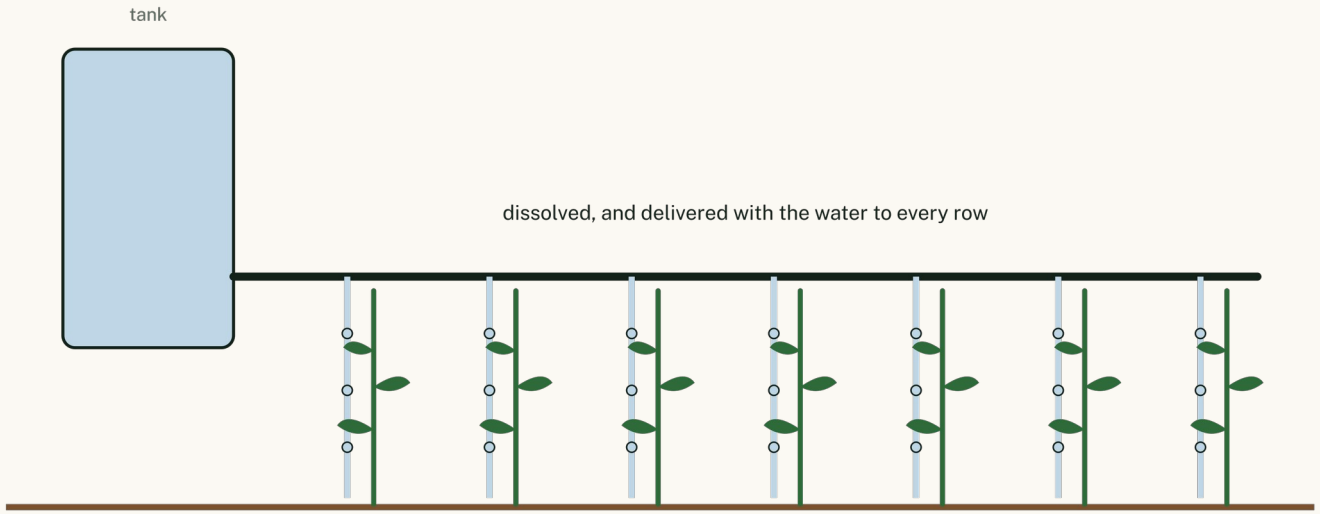
VAN's 2 single-zinc products are shown side by side, with prices from the VAN farmer price sheet, Aug 2026. The crop plans use them on different crops at different stages; the grower follows the plan for his crop.

Product	Pack	Zinc in the pack	Stage and method	Rupees
V-Zinc 10%, zinc sulfate, liquid	3 L	300 g	Early growth, in the irrigation water or on the leaf	1,375
V-Zinc 10%, the 20 L pack	20 L	2.0 kg	The same product, for many acres	6,450
V-Transform, zinc 21%, granular	3 kg	630 g	Mostly at germination, side-dressed or broadcast into the soil, in 20 crop plans	1,975

The dealer sells the product named in the crop plan; he does not sell by grams of zinc. The liquid goes in the irrigation water or on the leaf where the plan says fertigation or spray; the granule goes into the soil beside the crop row where the plan says side dressing. No crop plan uses both V-Transform and V-Zinc: V-Transform is in 20 plans, V-Zinc in 4. Some plans also carry VL-Micromix, which holds 5% zinc. Neither replaces the other for the same crop and stage.

Prices: the farmer price sheet, Aug 2026: V-Zinc 3 L 1,375 and 20 L 6,450; V-Transform 3 kg 1,975. Zinc per pack from the declared analyses (Zn 10% w/v; Zn 21% w/w). Stages and methods: the crop plans on van.com.pk.

Put it in the irrigation water, or spray it on the leaf. Never mix it with phosphate.



Fertigation: dissolve, then release slowly

On mango, the crop plans apply it in the irrigation water. Put it into the drum at the head of the watercourse or the drip tank, stir it in and let it run out slowly over the whole irrigation. It is fully soluble and leaves no residue in the tank or the emitter, as VAN's product data says.

Spray: water first, then the pack

Chickpea, lentil and mungbean take ⅕ of a pack, 600 ml, sprayed at early growth. Fill the tank first with 100 to 200 litres of clean water for each acre, then add the measured V-Zinc. Keep the tank agitated, and spray a fine, even fan in the morning or evening, never at midday. Use the plan's amount of V-Zinc in 100 to 200 litres of water for each acre; the rate is both amounts together.

Never in the same tank as a soluble phosphate

Zinc and a soluble phosphate make zinc phosphate, which does not dissolve: both are lost and the line can block. Give zinc its own spray or its own irrigation. Do not use these rates in a drone. Ask VAN for the drone rate.

OBJECTION 1 OF 7

“1,375 for 3 litres? A bag of zinc sulfate powder costs less.”

What the dealer says

Write down both prices. Then write down what each product is. This pack is 300 g of zinc already in solution, made to go through the drip line or the watercourse without residue, or onto the leaf with a jug’s measure, at the stage the crop plan gives. Zinc sulfate powder is a general product, and VAN gives no price for it. The crop plan gives a rate in litres at a set stage, and that rate is written for this pack.

The facts behind the answer

- V-Zinc 3 L is Rs 1,375; 20 L is Rs 6,450; 200 L is Rs 55,000 (farmer price sheet, Aug 2026).
- 100 g of zinc in every litre, 300 g in the 3 L pack (the declared analysis on the pack).
- Every V-Zinc row in the crop plans is written in the 3 L pack (the crop plans on van.com.pk).
- Fully soluble, no residue in the tank or the emitter (VAN’s product data).

Never say: “It gives you X more maunds”, a rupees-per-acre return, or a cost per gram of zinc against zinc sulfate powder. No yield figure exists for this product.

OBJECTION 2 OF 7

“Can I put it in the tank with my DAP water, or spray it with the phosphate?”

What the dealer says

No. Zinc and a soluble phosphate in 1 tank make zinc phosphate, which does not dissolve. You lose the zinc, you lose the phosphate, and the drip line can block. Give the zinc its own irrigation or its own spray. The same chemistry is why years of heavy phosphate on a zinc-short soil make the zinc shortage worse.

The facts behind the answer

- Zinc phosphate is poorly soluble: zinc and a soluble phosphate in 1 tank drop out together (textbook chemistry).
- Zinc and phosphorus interfere with each other in the soil; high pH locks zinc the same way it locks phosphate (VAN’s 17-nutrients page).
- Every V-Zinc row is applied on its own, by fertigation or spray; no row names another product for the same tank (the crop plans on van.com.pk).

Never say: “Stir it well and it is fine.”

OBJECTION 3 OF 7

“Is this the chelated zinc? I was told chelate is what works on our soil.”

What the dealer says

No. V-Zinc 10% is zinc sulfate, as the registered analysis says. VAN’s chelated zinc is a separate product for partners, and I will not sell you zinc sulfate as if it were the chelate. The crop plan applies the zinc sulfate as a solution: as a leaf spray on chickpea, lentil and mungbean and mash and in the irrigation water on mango. Either way it reaches the root zone or the leaf in a form the crop can take up. Follow the plan for your crop.

The facts behind the answer

- V-Zinc 10% is 10% zinc (w/v), supplied as zinc sulfate (the V-Zinc page on van.com.pk).
- VAN’s chelated zinc is a separate partner product with its own specification (VAN’s product data).
- V-Zinc rows: spray on 3 rows and fertigation on 1 (the crop plans on van.com.pk).

Never say: “It is as good as a chelate” or “it is a chelate”. Neither is on VAN’s record for this product.

OBJECTION 4 OF 7

“My crop is already showing it. Will a spray now fix it?”

What the dealer says

A spray reaches the leaf directly and is the quickest way to get zinc into a standing crop. If your crop’s plan has a spray row, apply that spray now. But the plant keeps the size it has already reached: zinc applied at flowering comes too late to change the plant’s size, because the shortage has already limited it. Next season, apply zinc at the stage the crop plan gives, before any sign of shortage appears. VAN gives no recovery figure and no number of days.

The facts behind the answer

- VAN names germination and early growth as the stages for V-Zinc (the V-Zinc page on van.com.pk).
- Zinc controls plant height, leaf size and, in rice and wheat, grain set (VAN’s 17-nutrients page).
- The spray rows are chickpea, lentil and mungbean at early growth, ⅕ pack (the crop plans on van.com.pk).

Never say: “It greens up in 3 days” or “it recovers the yield”.

OBJECTION 5 OF 7

“Which stage, and how much?”

What the dealer says

These are the crop plan rates, in 3 L packs: ⅕ pack by spray at early growth on chickpea, lentil and mungbean and mash; 2 packs by fertigation at early growth on mango. 20 other crop plans give zinc as V-Transform, the granular zinc, instead.

The facts behind the answer

- 4 crop plans name V-Zinc, in 4 rows, all in the 3 L pack (the crop plans on van.com.pk).
- The rates in the answer are the rows of the same plans.
- 20 crop plans use the granule, V-Transform, instead (the same plans).

Never say: “More packs, more yield”, or any rate for a crop that has no V-Zinc row.

OBJECTION 6 OF 7

“Can I put it in my drone?”

What the dealer says

Yes, but not at this rate. V-Zinc is 1 of 6 VAN brands approved for drone and ultra-low-volume spraying in its standard form, and van.com.pk says so. A drone tank holds about 20 litres, so the rate per tank is not the fertigation or knapsack rate. Ask VAN for the drone rate; I will not work one out from the crop plan rate.

The facts behind the answer

- V-Zinc is 1 of 6 VAN brands approved for drone and ultra-low-volume spraying in their standard form (the application-systems page on van.com.pk).
- A drone tank carries about 20 litres; everything the crop gets that day is dissolved or suspended in it (the same page).
- The fertigation and spray rows are written for the 3 L pack in the watercourse or the knapsack. Ask VAN for the drone rate per tank.

Never say or give: any drone rate worked out from the fertigation or spray rate.

OBJECTION 7 OF 7

“Is it registered? What is the licence?”

What the dealer says

Yes. V-Zinc 10% is registered under Pakistan Standard PS 5469-2020 for liquid nutrient supplements, and VAN makes it under manufacturing licence CM/L-3130/2024. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. The zinc test is run at VAN Lab but is outside its PNAC accreditation scope. On van.com.pk, enter the batch number printed on the pack to read the certificate for that batch.

The facts behind the answer

- V-Zinc 10%: PS 5469-2020, manufacturing licence CM/L-3130/2024 (the regulatory table on van.com.pk, from the PSQCA master record).
- The licence number is printed on the pack, and a dealer quotes that number.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017 (VAN’s quality page). The zinc test is run at VAN Lab but is outside its PNAC accreditation scope (the lab price list on van.com.pk).

Never say: “This licence is valid until …” Quote the licence number only; never state an expiry date for the licence.

ITS PLACE IN THE CROP PLANS

V-Zinc 10% in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Chickpea	Early Growth	⅕ pack	Spray	3 L
Lentil	Early Growth	1/5 pack	Spray	3 L
Mungbean & Mash	Early Growth	1/5 pack	Spray	3 L
Mango	Early Growth	2 packs	Fertigation	3 L

The rows are printed as published in the crop plans on van.com.pk. Coverage: these are all 4 rows in the 4 crop plans that name V-Zinc 10%. Of these 4 rows, 3 say spray and 1 says fertigation. Every row uses the 3 L pack. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield figure, a grain-zinc figure or a human-nutrition claim. No trial of V-Zinc exists. The published wheat response trials and the National Nutrition Survey are about zinc in general, and others carried them out, not VAN.

2. That 64% of Pakistan's soils, or 57% of Punjab's, are zinc-deficient: VAN has not named a source for either. The figure a dealer may give is from the Punjab soil survey: 59.0% of samples below 1.0 ppm zinc.

3. That it is chelated, or as good as a chelate. It is zinc sulfate, as registered; the chelate is a separate product.

4. That VAN sells a 1 L foliar pack with iron, amino acids and seaweed. It is not on the price sheet, and VAN has not confirmed it.

5. A drone rate, or a fertigation or spray rate that is not in a crop plan, as if it were a crop plan rate.

6. That it greens a crop in X days, or restores the growth of a plant whose size is already set.

HOW A GROWER VERIFIES A PACK

Every batch is tested, and the batch number on the pack identifies its certificate.

1 · The batch number

Every pack carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any pack by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for V-Zinc 10% shows.

The certificate gives the declared analysis tested for the batch: zinc against 10% w/v. The zinc test is run at VAN Lab but is outside its PNAC accreditation scope. The batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-3130/2024 are on the pack. The grower reads the certificate for his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Zinc is needed for the crop's height and leaf size. *English meaning · Urdu line pending review*	زنک فصل کے قد اور پتے کے پھیلاؤ کے لیے ضروری ہے
Your soil has zinc, but holds it in a form the root cannot take up. So give it as a liquid, in the water or on the leaf. *English meaning · Urdu line pending review*	زمین میں زنک ہے مگر ایسی شکل میں جو جڑ نہیں لے سکتی، اس لیے پانی میں یا پتے پر مائع دیں
Give it at early growth, while the plant's size is being set. *English meaning · Urdu line pending review*	ابتدائی بڑھوتری پر دیں، جب پودا اپنا قد طے کر رہا ہو
Never in the same tank as phosphate. *English meaning · Urdu reviewed 27 Sep 2026*	فاسفیٹ کے ساتھ ایک ٹینک میں کبھی نہیں

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

V-Zinc 10%: zinc in solution, in the irrigation water or on the leaf, early in the season, at the crop plan rate. Never mix it with phosphate.

